Graded Exercise 3: Hints and Tipps

Tipps and hints for the third graded exercise

Core Functionalities vs Bonus features

With a solid solution for the core featuews, you will the exercise with a decent grade. If you aim for 15 points (or more), you can address one or more of the bonus features. If you solve bonus features, this can earn you bonus points which we will add to previous exercise results if you did not achieve 15 points already.

Sub-tasks:

1. Analysis of Sequence Features

This requires a parser for the GFF file. The nine standard fields of GFF3 are:

1. seqid: The name of the chromosome, scaffold, or sequence.
2. source: The program, pipeline, or database that generated the annotation (e.g., Augustus or BestRefSeq).
3. type: The feature type (e.g., gene, mRNA, exon, CDS).
4. start: The 1-based start coordinate of the feature on the sequence.
5. end: The 1-based end coordinate of the feature.
6. score: A numerical score for the feature (often a floating-point value or . if not applicable).
7. strand: The orientation of the feature, represented as + (forward), - (reverse), or . (unstranded).
8. phase: For coding sequences (CDS), this defines where the feature begins relative to the reading frame (0, 1, or 2, or . for non-CDS).
9. attributes: Semicolon-separated key-value pairs containing extra metadata, such as ID (unique identifier) and Parent (to show hierarchy).

The tricky part is the parser for the attributes. This is key-value pairs splitted by semicolon. The DbXref key contains a further key value list, this time separated by comma (pairs) and colon (key:value). Have a look at the data, it hopefully becomes clear quite quickly.

2. Tabular Overview (CSV/TSV Export)

Think abaout how to store the data from subtask 1 into data structures and how to do that efficiently. You might think about a python class for GFF entries. This subtaks is mainly to create a new table with only the most important information from the GFF file. There are fields that are not found for all CDS, e.g. 'gene'. Take care this information is not lost.

3. Sequence Extraction

When extracting the sequenced from the genome fasta file, make sure you are on the correct positions. Remember that for CDS on the "-"-strand, you need to create reverse complementary sequences. Check the nucleotide sequences before converting to AA. For bacteria, ~90%+ should start with "ATG", if this is not the case, something is wrong in your nucleotide parser. For DNA->AA conversion, you have prepared code already during the lecture and exercises. Feel free to re-use.

4. Genome Analyses

- **Codon usage**: Which Codons (triplets) are used within all genes of all genomes. Sum up the data for all CDS and present it as a table with percentage values.
- **COG Categories**: Some CDS have more than one COG category, this looks like that: 'COG:OE". For such cases, please count both categories.

5. Visualization

Just be creative :)

Technical Requirements

- **Command Line Interface:** Use meaningful arguments. A good start: '-f fasta file, -g gff-file'.

General tipps

- **Create test data**: While coding, maybe you don't want to run your tests allways on the complete 4500 CDS. Create a subset, the first ~100 lines or so of the GFF files will work just fine, but will be much quicker for testing.
- **Look at runtimes**: Don't forget to keep an eye on runtimes, for a full genome, this task should be do doable with a few minutes of runtime. If you need considerably more time, look for inefficient code, e.g. unnecessary nested loops. If you struggle with runtime issues, feel always free to contact Stefan / Christopher / Jochen.

Optional Bonus Features

More ideas are welcome. Feel free to implement anything that seem useful to you.